

Sound of a star

Scientists have found a way to create 'sound' from a star's light and using it are able to decipher its size and stage of evolution

When are they going to admit it?

Now even NASA astronaut Chris Cassidy saw something floating around outside the ISS. Can the source get more credible? The world is ready to know.

Space Age

light signature of materials present and the third is a Neutral Mass Spectrometer that will measure the variance in the moon's atmosphere over several lunar orbits.

Mars Atmosphere and Volatile Evolution (MAVEN): Late 2013

NASA made huge news with the successful deployment of the Curiosity Rover onto the Red Planet. While the rover takes readings and samples from on-ground, NASA feels that some chemical analysis can only be performed from measurements made high in the atmosphere, which is where the MAVEN program comes in. MAVEN will orbit high above the surface so it can sample the upper atmosphere for signs of what changed over the eons and why. The mission will be the first of its kind and calls for instruments that can pinpoint trace amounts of chemicals high above Mars. The results are expected to let scientists test theories that the sun's energy slowly eroded nitrogen, carbon dioxide and water from the Martian atmosphere to leave it dry and possibly desolate.

Magnetospheric Multiscale Mission (MMS): 2014

The Magnetospheric Multiscale mission is a Solar Terrestrial Probes mission comprising of four identically instrumented probes that will use Earth's magnetosphere (the area around Earth that is effected by its magnetic field) as a laboratory to study the interaction between the Sun's and the Earth's magnetic fields and their effect on various particles.



LADEE – pronounced “laddie,” not “lady”

New Horizons: 2015

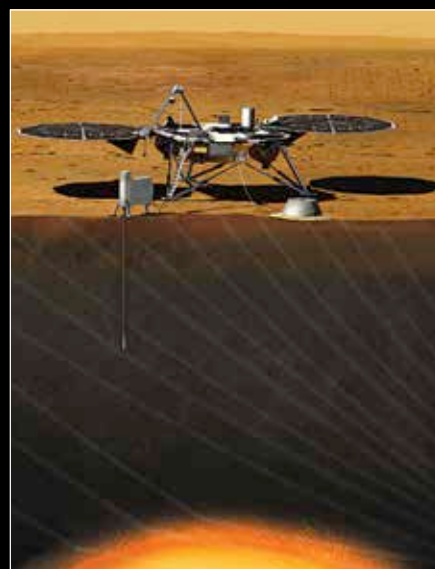
The New Horizons mission actually commenced back in 2006, when NASA launched spacecraft to Pluto. The probe has just passed the moons of Jupiter and will be the fifth to venture so far out away from our Sun. The New Horizons spacecraft will be performing a flyby mission of Pluto (and its moons) and will gather any observational data it can about what lies on the fringe of our solar system. NASA hopes to get a good glimpse and understanding of the former planet and its five known moons. NASA hopes to discover more with the New Horizons mission about the former planet and its surrounding moons. The spacecraft is expected to conduct its flyby sometime in 2015.

Juno: July 2016

No, this isn't the movie we're talking about there, but an unmanned spacecraft that is currently travelling at about 7 miles per second towards the largest planet in our solar system. The mission draws its name from Greek and Roman mythology, being named after Jupiter's wife. It was only Juno who was able to see through the veil of clouds Jupiter had cast around himself to hide his mischief. The mission is an attempt to take on that very nature, as Juno is set to arrive at Jupiter in July 2016 with the aim of understanding the giant's atmosphere. The spacecraft will also determine how much water is present in the planet's atmosphere to test the various theories of planet formation. It will also study the effect of Jupiter's magnetosphere along with deciphering the composition of the planet's atmosphere.

InSight: March 2016

Interior Exploration using Seismic Investigations, Geodesy and Heat Transport or InSight for short, is NASA's mission that seeks to understand the composition of the Red Planet. The mission will see NASA placing a single lander on Mars that will study its deep interior by drilling into the planet's surface. Since Mars lacks a tectonic plate system, NASA says the interior of the planet has a more accurate record of Mars's planetary evolution than the mountains and canyons we have been



InSight lander, launches in 2016

studying so far (as they are subject to environmental degradation and meteor and asteroid strikes).

Solar Probe Plus: July 2018

Apparently some scientists at NASA discovered energy signatures in the solar winds as they were picked up and measured by an aging probe, leading them to form new theories about nuclear fusion. To make more measurements, NASA will be launching the Solar Probe Plus towards our closest star, the sun. The probe will travel to the Sun's outer atmosphere, the farthest any spacecraft has ever gone. At its closest to the Sun, NASA says the spacecraft will be about 7 million kilometres from the star. Solar Probe Plus will have to withstand temperatures greater than 1400 Celsius and survive blasts of radiation at levels not experienced by any previous spacecraft. The probe will sample the sun's plasma and magnetic field at the very source of the solar wind.

Manned Mission to Mars: 2030

So far, all we have is President Obama's word on putting a man on Mars in the 2030's, the preparations for which are already underway. NASA says that within ten years of landing the second probe on Mars in 2020, they will put a man on the planet. Interestingly, Mars One, a privately funded venture, aims at putting the first two-member team on Mars by 2023. 